(1) (i) $\lim_{x \to a} \frac{x^{100} - a^{100}}{(x - a)}$

$100(x)^{100-1}$

$100x^{99}$

(ii) $\lim_{x \to (-2)} \frac{x^3 + 8}{x + 2}$

$\lim_{x \to (-2)} \frac{x^3 - (-2)^3}{x - (-2)}$

$3(-2)^2$

$3 \times 4 = 12$

(iii). $\lim_{x \to 2} \left(\frac{1/x^2 - 1/4}{(x-2)} \right)$

$\lim_{x \to 2} \frac{x^{-2} - 2^{-2}}{x - 2}$

$-2 (2)^{-3}$

$-2 \times \frac{1}{8}$

$\frac{-1}{4}$ //

(iv). $\lim_{x \to 1} \left(\dfrac{x^{3/4} - 1}{1 - x^{-2/3}} \right)$

$$-1 \quad \frac{\lim_{x \to 1} (x^{3/4} - 1)}{(x^{2/3} - 1)}$$

$$-1 \times 3/4$$

$$\lim_{x \to 1} \left(\frac{(x^{3/4} - 1)}{1 - x^{-2/3}} \right)$$

$$-1 \quad \lim_{x \to 1} \frac{(x^{3/4} - 1)}{(x^{-2/3} - 1)}$$

$$-1 \cdot \times \frac{3/4 (1)^{3/4 - 1}}{-2/3 (1)^{-2/3 - 1}} = \frac{9}{8}$$

$$-1 \times \frac{3}{4} \times -\frac{3}{2}$$

x

(v) $\lim_{x \to 1}$ $\cancel{x^2 - x}$ x^{-2}

(v) $$\lim_{x \to 1} \frac{x^2 - x^{-2}}{x^5 - x^{-5}}$$

$$\lim_{x \to 1} \frac{x^2 - \frac{1}{x^2}}{x^5 - \frac{1}{x^5}}$$

$$\lim_{x \to 1} \frac{\frac{x^4 - 1}{x^2}}{\frac{x^{10} - 1}{x^5}} \qquad \frac{f(x) \times g(x)}{}$$

$$\lim_{x \to 1} \frac{x^4 - 1}{\cancel{x^2}} \times \frac{\cancel{x^5}\, x^3}{x^{10} - 1}$$

$$\lim_{x \to 1} x^3 \times \lim_{x \to 1} \frac{x^4 - 1}{x^{10} - 1}$$

$$1 \times \frac{4 \times 1^3}{10 \times 1^9} = \frac{4}{10} = \frac{2}{5}$$

Asura

(2). (1)

$$\lim_{x \to 0} \frac{\tan x}{x}.$$

$$\lim_{x \to 0} \frac{\sin x}{\cos x \times x}.$$

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1}$$

$$\lim_{x \to 0} \frac{\sin x}{x} \times \lim_{x \to 0} \frac{1}{\cos x}.$$

$$= 1 \times 1$$

$$\underline{\underline{= 1}}$$

(2)

$$\lim_{x \to 0} \frac{2x - \sin x}{2x + \sin x}.$$

$$\lim_{x \to 0} \frac{\cancel{x}\left(2 - \frac{\sin x}{x}\right)}{\cancel{x}\left(2 + \frac{\sin x}{x}\right)}$$

$$\frac{\lim_{x \to 0} 2 - \lim_{x \to 0} \frac{\sin x}{x}}{\lim_{x \to 0} 2 + \lim_{x \to 0} \frac{\sin x}{x}}$$

$$\frac{2 - 1}{2 + 1} = \frac{1}{3} \,//$$

(iii) $\lim_{x \to 0} \frac{\sin 5x \cos 3x}{x}$

$5 \lim_{x \to 0} \frac{\sin 5x}{5x} \times \lim_{x \to 0} \cos 3x$

$5 \times 1 \times 1$

5 //

(iv) $\lim_{x \to 0} \frac{x + \sin 3x}{x - \tan 3x}$

$\lim_{x \to 0} \frac{\cancel{x}\left(1 + \frac{\sin 3x}{x}\right)}{\cancel{x}\left(1 - \frac{\sin 3x}{x \cos 3x}\right)}$

$\frac{\lim_{x \to 0} 1 + 3\lim_{x \to 0} \frac{\sin 3x}{3x}}{\lim_{x \to 0} 1 - \frac{3\lim_{x \to 0} \frac{\sin 3x}{3x}}{\lim_{x \to 0} \cos 3x}}$

$\frac{1 + 3}{1 - 3} = \frac{4}{-2} = -2$ //

(v). $\lim_{x \to 0} \frac{\tan 3x \sec 2x}{4x}$

$\lim_{x \to 0} \frac{\sin 3x}{4x \times \cos 3x \times \cos 2x}$

$\frac{3}{4} \lim_{x \to 0} \frac{\sin 3x}{3x} \times \lim_{x \to 0} \frac{1}{\cos 3x} \times \lim_{x \to 0} \frac{1}{\cos 2x}$

$\frac{3}{4} \times 1 \times 1 \times 1$

$\frac{3}{4}$ //

(vi) $\lim_{x \to \pi/4} \dfrac{\cot x - 1}{\cos 2x}$

$\cos 2x = \cos^2 x - \sin^2 x$

$\cos 2x = \cos^2 x - (1 - \cos^2 x)$

$= 2\cos^2 x - 1$

$\cos 2x = 1 - 2\sin^2 x.$

$$\lim_{x \to \pi/4} \frac{\frac{\cos x}{\sin x} - 1}{\left(\cos^2 x - \sin^2 x\right)}$$

$$\lim_{x \to \pi/4} \frac{\cancel{(\cos x - \sin x)}}{\sin x\,\cancel{(\cos x - \sin x)}\,(\cos x + \sin x)}$$

$$\lim_{x \to \pi/4} \frac{1}{\sin x} \times \lim_{x \to \pi/4} \frac{1}{\sqrt{2}\left(\frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x\right)}.$$

$$\frac{1}{\cancel{\sin \pi/4}} \times \frac{1}{\sqrt{2}} \lim_{x \to \pi/4} \frac{1}{\left(\sin\pi/4 \cos x + \cos\pi/4 \sin x\right)}.$$

$$\frac{1}{\cancel{\sqrt{2}}} \times \frac{\cancel{1}}{\cancel{\sqrt{2}}} \times \cancel{\sqrt{2}} \times \lim_{x \to \pi/4} \frac{1}{\sin\left(\pi/4 + x\right)}$$

$$\frac{1}{1} = 1$$

(vii) $\lim_{x \to 0} \frac{1 - \cos^2 2x}{x^2}$

$\lim_{x \to 0} \frac{\sin^2 2x}{4x^2} \times 4$

$4 \times \lim_{x \to 0} \frac{\sin^2 2x}{(2x)^2}$

$4 \times \left[\lim_{x \to 0} \frac{\sin 2x}{2x}\right]^2$

$4 \times 1 = 4$ //

(viii) $\lim_{x \to \pi/4} \frac{1 - \tan x}{\cos x - \sin x}$

$\lim_{x \to \pi/4} \frac{\cancel{(\cos x - \sin x)}}{\cancel{(\cos x - \sin x)} \cos x}$

$\lim_{x \to \pi/4} \frac{1}{\cos x}$

$\sqrt{2}$ //

(ix) $\lim_{x \to 0} \frac{1-\cos x}{x^2}$

$\cos 2x = 2\cos^2 x - 1$

$\cos 2x = 1 - 2\sin^2 x$.

$\lim_{x \to 0} \frac{\cancel{2}\sin^2 x/2}{\frac{x^2}{4} \times \cancel{4}\, 2}$

$\frac{1}{2}\left[\lim_{x \to 0} \frac{\sin x/2}{x/2}\right]^2$

$\frac{1}{2} \times 1 = \frac{1}{2}$.

(x) $\lim_{x \to 0} \frac{1-\cos 3x + 3x}{\sin 3x}$.

$\sin 2\theta = 2\sin\theta\cos\theta$

$\lim_{x \to 0} \frac{1-\cos 3x}{\sin 3x} + \lim_{x \to 0} \frac{3x}{\sin 3x}$.

$\lim_{x \to 0} \frac{\cancel{2}\sin^{\cancel{2}} 3x/2}{\cancel{2\sin 3x/2}\ \cos 3x/2} + \lim_{x \to 0} \frac{1}{\frac{\sin 3x}{3x}}$

$\lim_{x \to 0} \tan 3x/2 + \lim_{x \to 0} \frac{1}{\frac{\sin 3x}{3x}}$.

$0 + 1$

1

(xvi) $\lim_{x \to \pi/4} \dfrac{1-\sin 2x}{(\cos x-\sin x)(x-\pi/4)}$

$\lim_{x \to \pi/4} \dfrac{1-\cos(\pi/2-2x)}{(\cos x-\sin x)(x-\pi/4)}$

$\lim_{x \to \pi/4} \dfrac{2\sin^2(\pi/4-x)}{(x-\pi/4)\sqrt{2}\left(\frac{1}{\sqrt{2}}\cos x-\frac{1}{\sqrt{2}}\sin x\right)}$

$\lim_{x \to \pi/4} \dfrac{2\sin^2(\pi/4-x)}{(x-\pi/4)\sqrt{2}\left(\sin\pi/4\cos x-\cos\pi/4\sin x\right)}$

$\lim_{x \to \pi/4} \dfrac{2\sin^{\cancel{2}}(\pi/4-x)}{\sqrt{2}(x-\pi/4)\cancel{\sin(\pi/4-x)}}$

$-\sqrt{2}\lim_{x \to \pi/4} \dfrac{\sin(\pi/4-x)}{(\pi/4-x)} = -\sqrt{2}$.

Asura

ලවයේ බලය = හරයේ බලය
සමාන නම්
ලවයේ සං.
හරයේ සං.

(3) $\lim_{x \to \infty} \dfrac{ax^2 + bx + c}{lx^2 + mx + n}$

(ලවයේ බලය > හරයේ බලය) ∞

$$\lim_{x \to \infty} \frac{\cancel{x^2}\left(a + \frac{b}{x} + \frac{c}{x^2}\right)}{\cancel{x^2}\left(l + \frac{m}{x} + \frac{n}{x^2}\right)}$$

$$\lim \frac{a}{l} //$$

(4) (i) $\lim_{x \to \infty} \dfrac{x^4 + 3x + 3}{5x^2 + 1}$

$$\lim_{x \to \infty} \frac{\cancel{x^4}\left(1 + \frac{3}{x^3} + \frac{3}{x^4}\right)}{\cancel{x^2}\left(5 + \frac{1}{x^2}\right)}$$

$x^2 + 6x$

$$\frac{1}{5} //$$

(ii) $\lim_{x \to \infty} \dfrac{2x + 3}{x^2 + 5x - 1}$

$$\lim_{x \to \infty} \frac{x\left(2 + \frac{3}{x}\right)}{x^2\left(1 + \frac{5}{x} - \frac{1}{x^2}\right)}$$

$$\underline{0}$$

Date: __/__/__

(iii) $\lim_{x \to a} \frac{2x^3}{x^2}$

$2 \lim_{x \to \alpha} \frac{x^{\cancel{3}}}{\cancel{x^2}}$

$2 \times \alpha$

α

(iv) $\lim_{x \to \alpha} \frac{x^3 + 5x - 1}{2x^2 + 7}$

$\lim_{x \to \alpha} \frac{x^3 \left(1 + \frac{5}{x^2} - \frac{1}{x^3}\right)}{\cancel{x^2}\left(2 + \frac{7}{x^2}\right)}$

α

$$\lim_{x \to a} \frac{x^{?} + 3x + 3}{(5x^2 + 1)}$$

Lim

5 (i) $$\lim_{x \to 3} \frac{(x - 3)}{\cancel{(x+1)} \; (\sqrt{x+1} - 2)}$$

$$\lim_{x \to 3} \frac{(x - 3)}{(\sqrt{x+1} - 2)} \times \frac{(\sqrt{x+1} + 2)}{(\sqrt{x+1} + 2)}$$

$$\lim_{x \to 3} \frac{(x-3)(\sqrt{x+1} + 2)}{x + 1 - 4}$$

$$\lim_{x \to 3} \frac{\cancel{(x-3)}(\sqrt{x+1} + 2)}{\cancel{(x-3)}}$$

4 //